MILITARY SPECIFICATION SHEET

CONNECTORS, PLUGS, ELECTRICAL, COAXIAL, RADIO FREQUENCY,

(SERIES SMA (CABLED) - PLUG, PIN CONTACT, RIGHT ANGLE, CLASS 2)

This amendment forms a part of MIL-C-39012/56D, dated 3 October 1986, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 13

Delete and substitute the following note 1/.

"1/ The connectors that are in stock or distribution that were previously qualified and marked with the old part number shall also be considered acceptable for Government use until connector stock is purged. Applies to the category B part number change only (i.e., M39012/56- to M39012/56B)."

The attached insertable replacement pages listed below are replacements for stipulated pages. When the new pages have been entered in the document, insert the amendment as the cover sheet to the specification.

Replacement page	Pages replaced
3	3
4	4
5	5
6	6

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Army - EA, MI
Air Force - 11, 17, 99
NASA - NA
DLA - ES

User activities:
Army - AT, AV
Navy - AS, MC, OS, SH
Air Force - 19

Preparing activity:
Army - CR

Agent:
DLA - ES

(Project 5935-3618-01)

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.
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TABLE I. Dash numbers, cross reference and dimensions - Continued.

Dash number	Applicable cable	Dimensions	Inches-millimeters maximum 1/8/6/
Category D - Field replaceable - defined piece parts 2/4/5/3/			
3502 3602 7/ 4502 4602 7/	M17/60-RG142*@ M17/158-00001ø M17/128-RG400 M17/175-00001ø	A	1.375(34.93)

1/ Millimeters are in parentheses.

2/ These connectors are assembled, using the applicable crimp tool, to the specified cables stripped as shown on figure 3.

3/ These connectors have captivated center contacts.

4/ Complete connector assembly shall consist of a body, center contact, ferrule, and assembly instructions.

5/ Not for use in army equipment.

6/ Coupling nuts shall be corrosion resistant steel with a passivated finish per MIL-F-14072. (applies only to "-3XXX" SERIES CONNECTORS)

7/ No safety wire holes.

8/ For logistics purposes, only connectors with safety wire holes will be stocked.

9/ All corrosion resistant steel bodied connectors which are designed to be assembled to the outer conductor using solder shall be gold plated per MIL-G-45204, Type II, Class 1.

The latest version of each cable shall be applicable.

* Cable to be used when performing tests requiring cable except as in note @.

@ Cable to be used for the +200°C temperature cycling tests.

† Connectors mate with connectors of the same material; i.e., M39012/56-3001 mates with M39012/57-3001, and M39012/56-4001 mates with M39012/57-4001.

∞ Preferred die M22520/5-33 closure B, alternate die M22520/5-03 closure B.

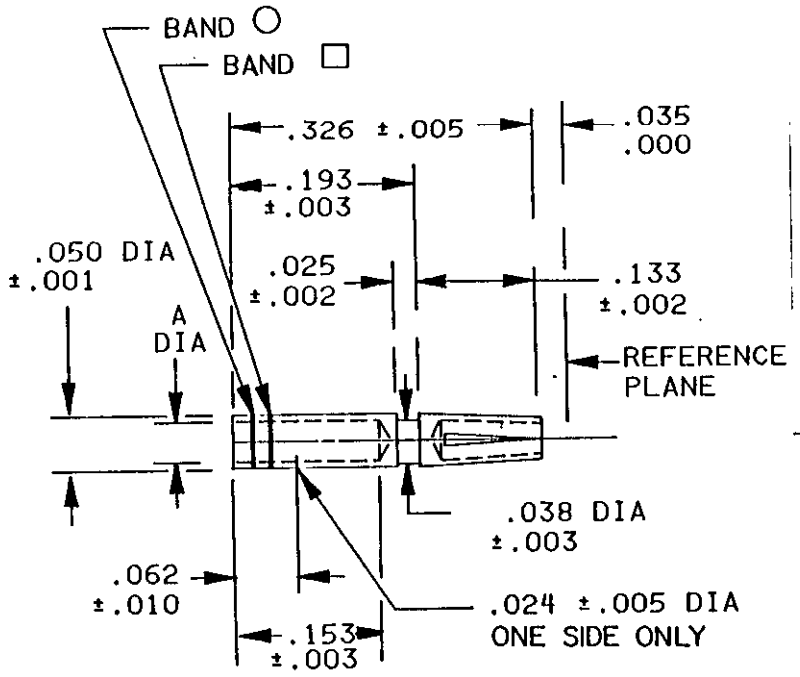
Σ Preferred die M22520/5-35 closure B, alternate die M22520/5-03 closure A.

& Preferred die M22520/5-41 closure B, alternate die M22520/5-05 closure B, or -09 closure A.

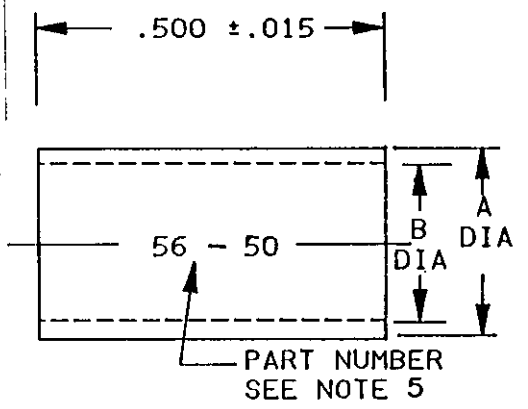
~ Preferred die M22520/5-19 closure B, alternate die M22520/5-05 closure A or -11, -57, closure A.

ø Caution is directed to the application of this cable above 400 MHz. Attenuation is tested only at 400 MHz. SRL and power handling capabilities are not stipulated herein.

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CENTER CONTACT



CRIMP FERRULE

Dash no.	Contact no. 2/	A ±.001	Basic crimp tool 1/	Crimp die or positioner	Crimp tensile min	Color band □	Color band ○
3502 4502	56-10	.041	M22520/1-01	Solder or M22520/1-15	4 lbs	Red	Maroon

Dash no.	Ferrule no. 2/	A ±.001	B ±.001	Basic crimp tool 1/	Crimp die or positioner
3502 4502	56-50	.250	.220	M22520/5-01	M22520/5-05, 11, 57 Closure A or 19 Closure B

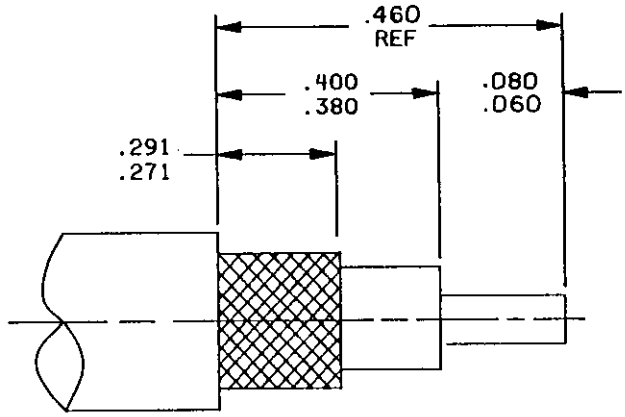
Inches	mm	Inches	mm
.001	0.03	.041	1.27
.002	0.05	.050	1.57
.003	0.08	.062	2.54
.005	0.13	.153	3.89
.010	0.25	.193	4.90
.015	0.38	.220	5.59
.024	0.61	.250	6.35
.025	0.64	.326	8.28
.035	0.89	.500	12.70
.038	0.97		

1/ Class 2 tool may be used by OEM (see MIL-C-22520).
2/ Contact numbers and ferrule numbers are for identification only.

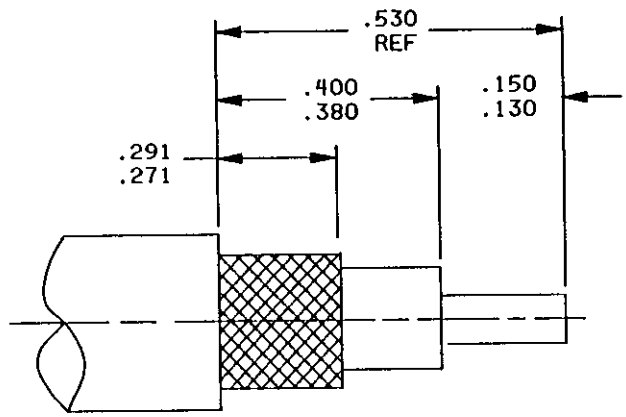
NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Solder or crimp tensile test shall be in accordance with MIL-C-39029.
4. Copyright notice: All information disclosed in these specification sheets which is or may be copyrighted is reproduced herein with the express permission of the copyright owner.
5. Contact numbers and ferrule numbers are for identification only.

FIGURE 2. Contact and ferrule dimensions for category D only.

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For category C



For category D

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

Inches	mm
.060	1.52
.080	2.03
.130	3.30
.150	3.81
.271	6.88
.291	7.39
.380	9.65
.400	10.16
.460	11.68
.530	13.46

FIGURE 3. Cable stripping dimensions for field replaceable connectors.

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ENGINEERING DATA:

Nominal impedance: 50 ohms.

Frequency range: 0 to 12,400 MHz.

Voltage rating: The voltage rating shall be in accordance with table II.

TABLE II. Voltage rating.

Cables	Voltage max (at sea level)	Voltage max (70,000 ft)
	V rms	V rms
M17/93-RG178, M17/169-00001	170	45
M17/54-RG122, M17/157-00001 M17/119-RG174, M17/173-00001 M17/94-RG179 M17/113-RG316, M17/172-00001 M17/152-00001	250	65
M17/28-RG058, M17/155-00001 M17/60-RG142, M17/158-00001 M17/84-RG223, M17/167-00001 M17/111-RG303, M17/170-00001 M17/128-RG400, M17/175-00001	335	85

Temperature rating: -65°C to +165°C.

REQUIREMENTS:

Dimensions and configuration: See figure 1.

Force to engage and disengage:

Longitudinal force: Not applicable.

Torque: 2 inch-pounds, maximum.

Coupling proof torque: 15 inch-pounds, minimum.

Safety wire hole tearout: 15 pounds, minimum.

Inspection conditions: For each test of threaded coupling connector where the test is performed on mated pairs, the pairs shall be torqued to 7 to 10 inch-pounds.

Mating characteristics: Reference MIL-STD-348 for dimensions.

Hermetic seal: Not applicable.

Leakage (pressurized connectors): Not applicable.

Insulation resistance: Method 302 of MIL-STD-202, test condition B.

5,000 megohms, minimum.

Center contact retention: 6 pounds, minimum axial force. Applicable to captivated-center-contact connectors only.

Radial torque: Not applicable.

Corrosion (salt spray): Method 101 of MIL-STD-202, test condition B.

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